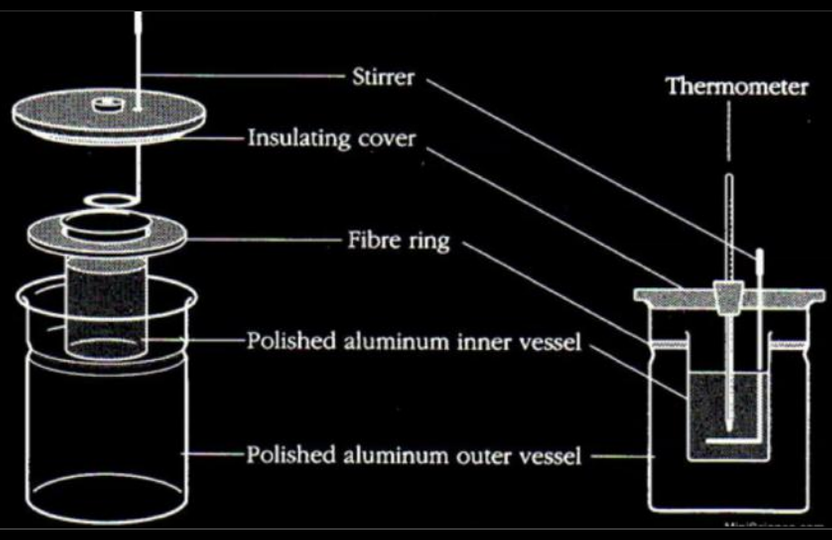
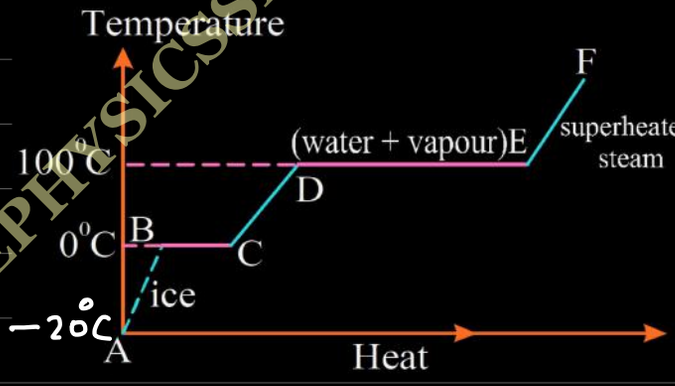
CALORIMETER: It is an apparatus used for doing several thermodynamics experiments involving heating of substances or mixing of substances. As shown in the figure, essentially a calorimeter has a well insulated container with a stirrer for mixing and provision for inserting thermometer for measuring temperatures of substances.



HEATING CURVE OF WATER:

Consider a block of ice say initially at $-20^\circ C$. As heat is supplied to the piece, the temperature begins to rise until it becomes $0^\circ C$ (see adjacent figure). Upon further heating it is seen that temperature remains constant for a while and during this time the ice keeps melting. (BC portion of the curve), the heat supplied during this time is called latent heat of fusion. After this the temperature once again begins to rise (liquid water begins to get hotter) and temperature keeps rising until it becomes $100^\circ C$. Once again the temperature becomes constant and water keeps boiling (passing in to gaseous phase) until entire water is converted to steam (DE portion of the curve), the heat supplied for this duration is called latent heat of Vaporization. On further heating, the temperature once again begins to rise.



→ Slope of section AB is double that of CD (for ice we need 0.5 cal/g°C (i.e 0.5 calorie of heat is required for producing 1°C temperature per gram), while for heating liquid we need 1 cal/g°C.

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SPECIFIC LATENT HEAT: Heat required per unit mass of a substance for changing its phase is called specific latent heat.

→ Specific latent heat of fusion: Solid → Liquid
→ Specific latent heat of vaporization: Liquid → Gas
→ Specific latent heat of sublimation: Solid → Gas.

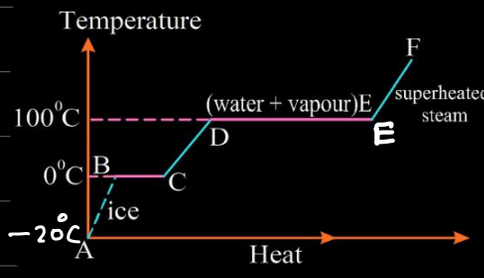
If Q heat is required for phase change of m mass of substance then specific latent heat:

$$L = Q/m \Rightarrow \boxed{Q = mL}$$

For ice specific latent heat of fusion: $\boxed{L_f = 80\ cal/g}$

For water to steam latent heat of vaporisation: $\boxed{L_v = 540\ cal/g.}$

SPECIFIC HEAT: The heat required per unit mass per unit rise in temperature is called specific heat of substance. Mathematically if ΔQ heat is required for m mass of substance for ΔT rise in temperature then specific heat is given as:



$$S = \Delta Q/m\Delta T \Rightarrow \boxed{\Delta Q = mS\Delta T}$$

If specific heat is not constant, then total heat supplied can be found by integration:

$$\Delta Q = \int mS\,dT$$

Here S will be a known function of temperature.

Having defined specific heat and latent heats, we can infer the following from the heating curve:

→ Specific heat of ice $= \dfrac{1}{m \times \text{Slope of } AB}$ $\left(\because S = \dfrac{1}{m}\dfrac{dQ}{dT}\right)$

→ Specific heat of water (liquid) $= \dfrac{1}{m \times \text{Slope of } CD}$

→ Specific latent heat of fusion $= (\text{Length of } BC)/m$

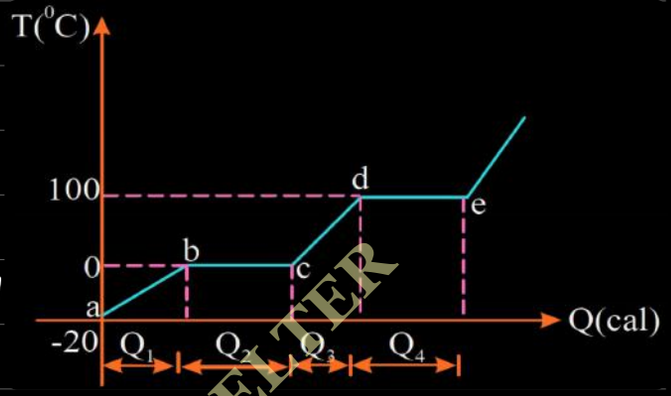
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→ Specific latent heat of vaporisation = (length of DE)/m.

→ Specific heat of water (liquid): **1 cal/g/°C**

→ Specific heat of ice = **0.5 Cal/g/°C**

EXAMPLE: Following Graph Represents Change in state of 10g of ice at −20°C. Find the net heat required to convert ice in to steam at 100°C.



Sol: $Q_1 = mS_s\Delta T = 10 \times 0.5 \times 20 = 100$ Cal

$Q_2 = mL_f = 10 \times 80 = 800$ Cal

$Q_3 = mS_l\Delta T = 10 \times 1 \times 100 = 1000$ Cal

$Q_4 = mL_v = 10 \times 540 = 5400$ Cal.

Total $Q_1 + Q_2 + Q_3 + Q_4 = 7300$ Cal

→ Note: We assumed specific heat of water to be constant at all temperatures in liquid phase, though it is not strictly true.

DEFINITION OF CALORIE: Amount of heat needed to increase temperature of 1g of water from 14.5°C to 15.5°C at a pressure of 1atm is called 1 Calorie. **1 Cal = 4.186 J**

PRINCIPLE OF CALORIMETRY: Total heat given by hot objects is equal to total heat gained by cold objects. If we treat heat like signed scalar (consided heat gained as positive) then:

$$\sum \Delta Q_i = 0 \qquad \Delta Q_i = \text{Heat gained by } i^{th} \text{ Substan}$$

HEAT CAPACITY OF A BODY (or amount of Substance): Amount of heat needed per unit rise in temperature of a body (or amount of substance) is called heat capacity. Mathematically:

$$H = \frac{Q}{\Delta T} = \frac{mS\Delta T}{\Delta T} = mS$$

$$\Rightarrow \boxed{H = mS}$$

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Thermal Equilibrium Of Non Reacting Substances Without Phase Change

Suppose we bring n non reacting substances in to thermal contact through mixing or otherwise then, by Principle of calorimetry:

$$\sum \Delta Q_i = 0$$

$$\Rightarrow \sum m_i S_i \Delta T_i = 0$$ Now let initial temperatures be T_i then:

$$\sum m_i S_i (T - T_i) = 0 \Rightarrow \boxed{T = \frac{\sum m_i S_i T_i}{\sum m_i S_i}}$$

[Here it is assumed that during thermal contact volume of the system remains conserved, otherwise we will also have to take in to account work done by the system].

Mixing Of Water And Ice: When a quantity of ice is brought in thermal contact with water the final state of mixture will depend on quantities and temperatures of ice and water. The possibilities are:

1. Entire water freezes
2. Entire ice melts
3. Some water freeze and finally we have a mixture of ice and water at 0°C
4. Some ice melts and finally we have a mixture of ice and water at 0°C.

To decide the final outcome, we can calculate the following Quantities:

Q_{SS} → Heat absorbed by ice to get to 0°C

Q_{LL} → Heat given by water to get to 0°C

If $Q_{SS} > Q_{LL}$ then atleast some water will freeze even if not completely, [Rules out entire liquid possibility]

If $Q_{LL} > Q_{SS}$ ⇒ Some ice will melt even if not entirely. [Rules out entire solid possibility].

Apart from this we can also calculate quantities:

Q_{SL} → Heat absorbed by ice at 0°C to melt completely

Q_{LS} → Heat given by water at 0°C to freeze completely at 0°C.

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Now if $Q_{SS} \geq Q_{LL} + Q_{LS}$ $\Rightarrow$ Entire mixture freezes

if $Q_{LL} \geq Q_{SS} + Q_{SL}$ $\Rightarrow$ Entire mixture melts

if $Q_{SS} > Q_{LL}$ but $Q_{SS} < Q_{LL} + Q_{LS}$ $\Rightarrow$ Liquid Partly freezes

if $Q_{LL} > Q_{SS}$ but $Q_{LL} < Q_{SS} + Q_{SL}$ $\Rightarrow$ Solid partly melts.

EXAMPLE: A piece of ice of mass 100 g and at temperature 0°C is put in 200 g of water at 25°C. What will be the final state of mixture.

Solution: $Q_{SL} = 100 \times 80 = 8000$ Cal.

$Q_{LL} = 200 \times 1 \times 25 = 5000$ Cal

$Q_{LL} < Q_{SL}$ $\Rightarrow$ Heat given by water is not sufficient to melt entire ice $\Rightarrow$ partly ice will melt (that can be melted using 5000 Cal)

Let m_1 be the mass of ice that melts $\Rightarrow$

$$m_1 L = Q_{LL} \Rightarrow m_1 \times 80 = 5000$$

$$\Rightarrow m_1 = 5000/80 = 62.5 \text{ g Ans.}$$

EXAMPLE: 2 Kg of ice at −20°C is mixed with 5 Kg of water at 20°C in an insulating vessel having negligible heat capacity, assuming no heat loss to surroundings what will be final state of the mixture?

Solution: $Q_{SS} = 2000 \times 0.5 \times 20 = 20000$ Cal

$Q_{LL} = 5000 \times 1 \times 20 = 100000$ Cal

$Q_{LL} > Q_{SL}$ $\Rightarrow$ atleast some ice must melt

Now $Q_{SL} = 2000 \times 80 = 160000$ Cal

$Q_{LL} < Q_{SS} + Q_{SL}$ $\Rightarrow$ Ice partly melts.

Now $Q_{LL} - Q_{SS} = 100000 - 20000 = 80,000$ Cal. This much heat is available for melting ice. Let m_1 be the mass of ice melted

$\Rightarrow$ $m_1 \times 80 = 80,000$ $\Rightarrow m_1 = 1000$ g $\Rightarrow$ 1 Kg ice melts

$\Rightarrow$ in final mixture:

Liquid = 5 Kg + 1 Kg = 6 Kg

Ice = 2 Kg − 1 Kg = 1 Kg.

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→ A similar logic can be built when steam is brought in thermal contact with ice. There we will have to calculate quantities like Q_{AA}, Q_{AL}, Q_{LA} etc also.

EXAMPLE: 20g of steam at 100°C is passed in to 100g of ice at 0°C. Find final state of mixture if latent heat of steam is 540 Cal/g.

Solution: Here final state could be one of the three depending on amount of steam of ice:

① ice + Water
② steam + Water
③ Entire water.

Now $Q_{VL} = 20 \times 540 = 10800$ Cal

$Q_{SL} = 100 \times 80 = 8000$ Cal

Now let us assume final state to be water at temperature T (valid only if T comes between 0°C & 100°C)

Now heat gained by ice = heat lost by steam

$\Rightarrow 8000 + 100 \times 1 \times T = 10,800 + 20 \times (100 - T)$

$\Rightarrow 120T = 10,800 + 2000 - 8000 = 4800 \Rightarrow T = 40°C.$

Thus final state is pure water at 40°C.

WATER EQUIVALENT OF A CALORIMETER: Amount of water which has same heat capacity as a given calorimeter is called water equivalent of calorimeter, (Or the amount of water which will absorb same amount of heat as given a given calorimeter for same rise in temperature).

For ease of calculations if liquid water does not freeze or vaporise then water equivalent of calorimeter can be simply be added to water mass.

CAUTION: If water freezes or vaporises, we can't simply add water equivalent of calorimeter to waters because obvious calorimeter is not undergoing phase change unlike liquid water, However no issue will arise if steam or ice is getting converted to liquid).

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EXAMPLE: Steam at 100°C is passed in to 1Kg of water at 9°C contained in a calorimeter of water equivalent 0.2Kg until the temperature of calorimeter and water in it rises to 90°C. Find the mass of steam condensed.

Solution: Heat gained by Water + Calorimeter = Heat lost by steam.
Let m_v be the amount of steam condensed, then:

$$(1000+200) \times 1 \times (90-9) = m_v \times 540 + m_v \times 1 (100-90)$$

$$\Rightarrow 550 m_v = 97200 \Rightarrow m_v = 176.7\,g$$

(Note since all calculations are done in calories and grams so we are a bit casual about writing units, though strictly it should be written).

SUPERCOOLED WATER: If a very pure sample of water is cooled, it is possible to maintain a liquid phase even at temperatures less than 0°C, such water is called supercooled water.
(This happens because there are no impurities which serve as nucleation sites for crystallisation to start). If supercooled water is shaken or some impurity is added, the whole sample immediately freezes in to ice.

WORDS OF CAUTION (Dealing with Supercooling):

→ Latent heat of freezing 80 cal/g is defined only at 0°C and can't be applied at any other temperature below 0°C
∴ for purpose of calculating final temperatures involving freezing of supercooled water, imagine first adding heat to bring it to water at 0°C and then subtract the added heat to compensate.

→ Specific heat of water is taken as 1 cal/g°C even though temperature is less than 0°C.

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EXAMPLE: What fraction of water supercooled to $-20^\circ C$ under standard pressure turns in to ice under standard pressure?

Solution: Consider 1gm of sample for convenience:

$Q_{LL} = \underset{m}{1} \times \underset{s}{1} \times \underset{\Delta T}{20} = 20$ Cal. (Converting Super cooled liquid to liquid at $0^\circ C$).

Now We shall subtract 20 Cal heat from water at $0^\circ C$. Let m be the mass of ice at $0^\circ C$.

$\Rightarrow \quad 20 = m \times 80 \Rightarrow m = 1/4$ gm.

$\Rightarrow$ 25% of supercooled water will become ice and 75% will still be liquid and finally we have a mixture of ice & water.

We can also find this fraction for general temperature. Let initial temperature be $-T^\circ C$.

$$\Rightarrow Q_{LL} = mcT = \eta m L$$

where η is fraction frozen.

$$\Rightarrow \eta = \left|\frac{cT}{L}\right| = \frac{|T|}{80}$$

EXAMPLE: At what initial temperature of supercooled water will entire supercooled water freeze.

Sol: $\eta = \frac{|T|}{80}$ For completely freezing, $\eta = 1 \Rightarrow T = -80^\circ C$.

SUPERHEATED WATER: When even after increasing the temperature beyond $100^\circ C$, the liquid phase is retained, it is known as superheated water. (This can be achieved by increasing pressure).

For superheated water also we take specific heat as 1 cal/g/°c for SEE Purposes)

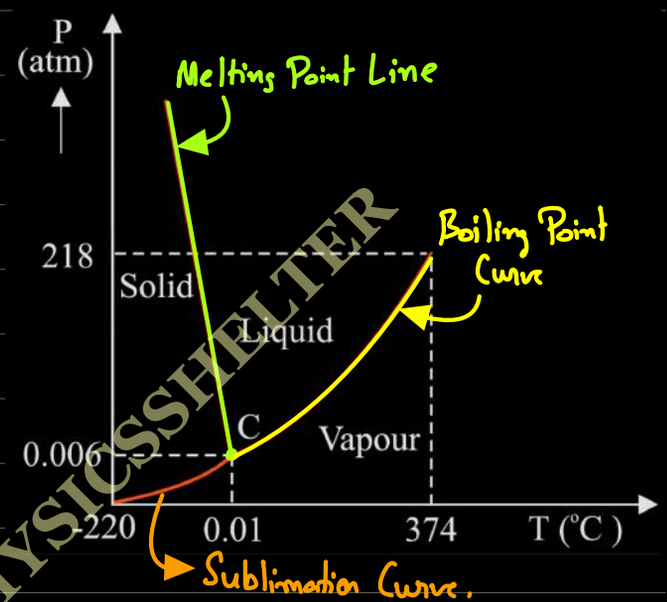
For Steam phase specific heat is about 0.5 cal/g/°c (0.48 to second decimal).

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Pressure Temperature Phase Diagrams:

A diagram representing phase of a substance at various pressures and temperatures is called Pressure, Temperature phase diagram. It helps in finding boiling points, melting points and sublimation points.

Pressure Temperature Phase Diagram For Water:



Triple Point: The point (P,T) where all the three phases (solid, liquid and gas) can coexist is called triple point, for water this is (0.006 atm, 0.01°C).

Critical Temperature: The temperature beyond which it is not possible to liquefy a substance by increasing pressure by any amount is called critical temperature. For water this is 374°C.

Elevation In Boiling Point: As can be readily seen from yellow curve in the phase diagram, by increasing pressure, boiling point increases. The increase in boiling point can be calculated using Clausius Clapeyron Equation.

Clausius Clapeyron Equation: This equation tells us the effect of increasing pressure on melting and boiling points. The equation is as follows:

$$\frac{dP}{dT} = \frac{L}{T(V_f - V_i)}$$

$L \longrightarrow$ Specific Latent Heat (of fusion or vaporisation)

$T \longrightarrow$ Temperature (Melting point or boiling point)

$V_f \longrightarrow$ Final specific Volume (liquid phase for melting, Vapour phase for boiling)

$V_i \longrightarrow$ Initial specific Volume. (solid Volume for melting, liquid Volume for boiling)

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$dP \rightarrow$ Change in Pressure.

$dT \rightarrow$ Change in temperature required for phase change.

DEPRESSION IN MELTING POINT: For water, volume in liquid phase is smaller than in solid phase, thus $V_f < V_i$, according to clausius clapeyron equation, if pressure is increased i.e. $dP > 0$ then $dT < 0$ or melting point must decrease.

REGELATION: When we have ice at $0^\circ C$, it will melt upon application of pressure according to clausius Clapeyron equation, however when pressure is removed, it will freeze again. This helps in skiing (also formation of "Golas", "Kalakhatta" Chuski etc) Due to decrease in melting point, ice under skis melts at the same temperature thus providing natural lubrication for sliding of skis.

EXAMPLE: The melting point of ice is $0^\circ C$ at 1 atm. At what pressure will it be $-1^\circ C$.

Solution: As per clausius clapeyron equation: $\frac{\Delta P}{\Delta T} = \frac{L}{T(V_f - V_i)}$

$\Rightarrow \Delta P = \frac{\Delta T}{T} \times \frac{L}{(V_f - V_i)}$ Now $L = 80 \times 4.2$ J/g

$V_f = 1\ cm^3/g$, $V_i = \frac{1}{0.9}\ cm^3/g$.

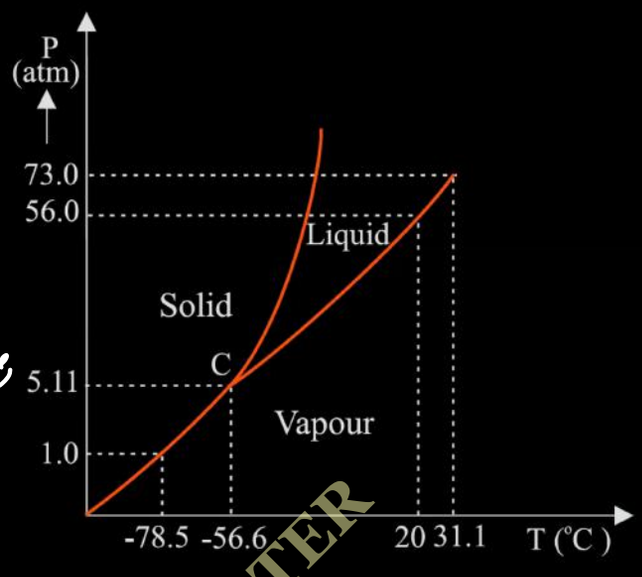
$\Delta T = -1$

$\Rightarrow \Delta P = \frac{-1}{273} \times \frac{80 \times 4.2\ (J/g)}{(1 - 1/0.9)\ (cm^3/g)} = 11.08\ J/cm^3 = 110.8$ atm.

$\Rightarrow$ Total pressure $= P_0 + \Delta P = 111.8$ atm Ans.

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Pressure Temperature Phase Diagram For Carbon Dioxide.



Triple Point: $(5.11 atm, -56.6°C)$
Critical Temp: $31.1°C$.

It can be readily seen that at atmospheric pressure, and room temperature, CO_2 is a gas whereas at atmospheric pressure, and very low temperatures it is solid. ∴ Solid CO_2 is called dry ice (Sublimates without producing liquid).
∴ Dry ice is used for stage shows (Producing fog/smoke effect).

Example: You have $m_w = 1.0 Kg$ of supercooled water at temperature $\theta_w = -10°C$ kept in a container and crushed ice at $\theta_i = -20°C$ kept in another container. How much of this ice must be added so that whole water freezes? Ignore losses and heat capacities of vessels.
Melting Point of ice = $\theta_0 = 0°C$.

Solution: Heat gained by ice = Heat lost by water

→ Remember any accounting of supercooled water involving phase change must be routed through $0°C$ because latent heat is defined at $0°C$ only.

Heat lost by water $= -m_w s_w (\theta_0 - \theta_w) + m_w L$
Heat gained by ice $= m_i s_i (\theta_0 - \theta_i)$
Equating:

$$-m_w s_w (\theta_0 - \theta_w) + m_w L = m_i s_i (\theta_0 - \theta_i)$$

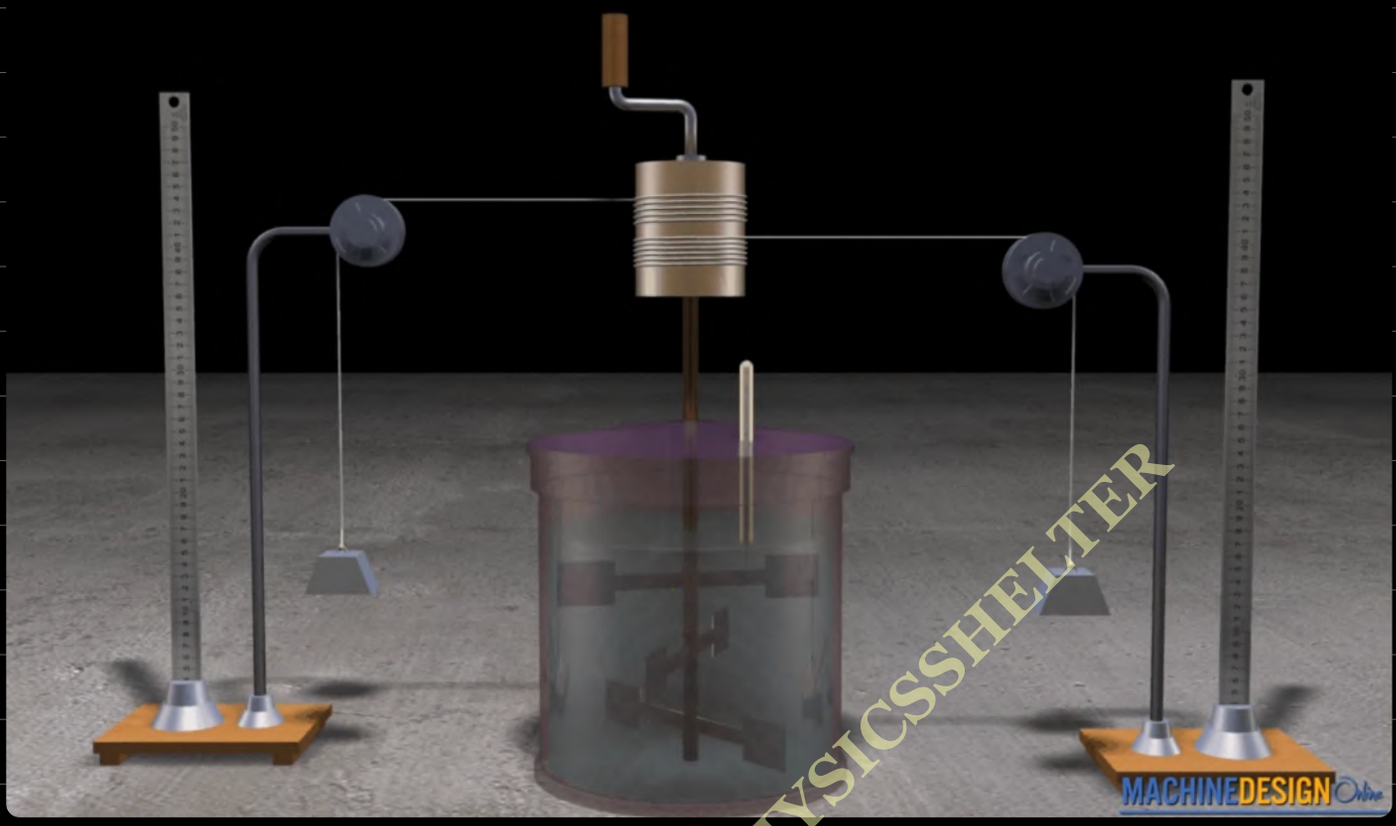
$$\Rightarrow m_i = \frac{m_w (L - s_w(\theta_0 - \theta_w))}{s_i (\theta_0 - \theta_i)}$$

Putting Values:

$$m_i = \frac{1000(80 - 1 \times 10)}{0.5 \times 20} = 7000 g = 7 Kg \text{ Ans.}$$

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Joule's Experiment For Finding Mechanical Equivalent Of Heat



In earlier days, heat was not recognised as a form of energy, Heat was something that was required to raise the temperature of a body. Calories was defined as unit of heat.

Joule showed that just as temperature could be increased by supplying heat, it could also be increased by doing mechanical work. In the above apparatus, the mechanical energy lost by falling weights was used to raise temperature of water through spinning paddles. Thus it was established that:

$W = JH$ where H is the heat required to raise the temperature of water by same amount as supplying W amount of Work. If W and H were measured in same units, $J = 1$, however, Heat was measured in calories and Work was measured in Joules, $\therefore J = 4.186$. ($\because$ 1 Joule = 4.186 Cal).

Boiling Vs Evaporation: Boiling occurs at a specific temperature when entire liquid is at the verge of phase change. (That's why you see bubbles at the bottom of container on burner when boiling happens. Whereas evaporation is surface phenomenon occuring at

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every temperature. [Energetic molecules escape from surface due to their random kinetic energy].

EXAMPLE :

A thermally insulated vessel contains some water at 0^0C. The vessel is connected to a vacuum pump to pump out water vapour. This results in some water getting frozen. It is given Latent heat of vaporization of water at 0°C = 21×10^5 J/kg and latent heat of freezing of water = 3.36×10^5 J/kg. The maximum percentage amount of water that will be solidified in this manner will be :-

(A) 86.2% (B) 33.6% (C) 21% (D) 24.36%

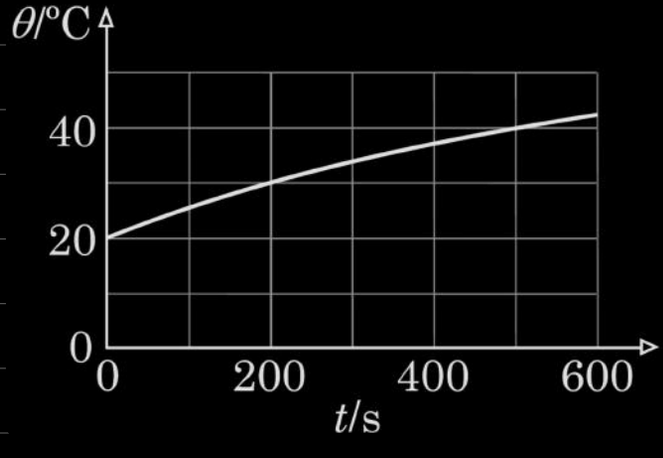
Let us consider 1Kg of water out of which x Kg solidifies and $(1-x)$ Kg evaporates. Now heat lost by x Kg = Heat Gained by $(1-x)$ Kg $\Rightarrow$

$$x \times 3.36 \times 10^5 = (1-x) \times 21 \times 10^5 \Rightarrow x = 0.862$$

$\Rightarrow$ 86.2% water freezes. (A).

PATHFINDER EXAMPLE: (BUILD-2)

Hot water from a reservoir maintained at constant temperature is being added at a very slow and constant rate in a calorimeter initially containing 1.0 kg of water at temperature 20 °C. The water in the calorimeter is being continuously stirred to maintain its temperature uniform. The graph shows how the temperature θ of water in calorimeter changes with time t. Assuming heat loss to the surroundings and work done in the stirring process to be negligibly small, find the temperature of the hot water reservoir.



Solution: Let rate of addition of hot water be λ Kg/sec. Upon mixing hot water and cold water, the temperature achieved is weighted average of their temperatures. Now we have two convenient data points to make the equations based on calorimetry principle i.e. (200, 30) and (500, 40)

Now let temperature of hot water be Θ then:

$$\frac{1 \times 10 + \lambda \times 200 \times \theta}{1 + 200\lambda} = 30 \text{ (1) and } \frac{1 \times 10 + \lambda \times 500 \times \theta}{1 + 500\lambda} = 40 \text{ (2)}$$

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Solving ① & ②, we get: $\lambda = \frac{1}{1000}$ kg/sec = 1 g/sec & $\theta = 80°C$ Ans.

PATHFINDER EXAMPLE: (Build 9)

A mixture of water and ice is kept in a well-insulated calorimeter of negligible heat capacity. A heater, which can supply heat at a constant rate 50 W to the mixture, is switched on. At the ends of second, third and fourth minutes after the heater is switched on, temperature of the mixture becomes 0 °C, 2 °C and 7 °C respectively. How many grams of water and ice were initially present in the calorimeter?

Specific latent heat of melting of ice is 390 J/g and specific heat of water is 4.0 J/(g·°C).

Solution: Let mass of water be x and that of ice be y. Initial temperature must be 0°C because both phases coexist and possibly continue to co-exist till end of 2 minutes. (∵ temperature at end of 2 min is still 0°C).

Now heat supplied till end of third minute:

$$Q_3 = 50 \times 60 \times 3 = 9000\ J,$$ this must be equal to heat gained by water and ice to achieve 2°C.

$$\Rightarrow x \times 4 \times 2 + y \times 390 + y \times 4 \times 2 \Rightarrow 8x + 398y = 9000 \quad (1)$$

Similarly by end of 4th minute, $Q_4 = 50 \times 60 \times 4 = 12000\ J$

$$\Rightarrow x \times 4 \times 7 + y \times 390 + y \times 4 \times 7 = 12000 \Rightarrow 28x + 418y = 12000 \quad (2)$$

From ① & ②:

$$x = 130\ g,\ y = 20\ g$$

⇒ Mass of Water = 130 g.

Mass of Ice = 20 g Ans.

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